

HIERARCHICAL BAYES ANALYSIS OF THE SIMPLEST CASE-CROSSOVER DESIGN

*Lize van der Merwe & C.J. Lombard**

Department of Statistics and Actuarial Science, University of Stellenbosch,
Private Bag X1, Matieland, 7602, South Africa.
e-mail: *lvdm@akad.sun.ac.za*.

*Biostatistics Unit, National Program for Health Systems and Policy
Research, Medical Research Council, P.O. Box 19070, Tygerberg, 7505,
South Africa.

e-mail: *Carl.Lombard@mrc.ac.za*.

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Summary: In the simplest form of a case-crossover design, the relative risk of an event (such as death) under two different ‘exposures’ is assessed. The exposure is observed only twice, once at the time of the event. The probability of exposure is required for each individual in the study. These probabilities are traditionally estimated using historical data. The traditional approach yields probabilities of 0, $\frac{1}{2}$, and 1 only.

A hierarchical Bayesian model is used to obtain more realistic estimates of these probabilities and consequently of the relative risk. Differences between the conventional and Bayes results are exhibited in specific examples.

1. Introduction

In a case-crossover design the relative risk of an event under different levels of exposure is assessed. Examples are

- the relative risk of a heart attack, given a certain level of exercise;
- the relative risk of pain relief, given a certain dose of an analgesic.

For each case, the exposure at the time of the event is observed. The 'usual' exposure of the case in the period prior to the event is also determined. The relative risk is the ratio:

$$\frac{P(\text{event}|\text{specific exposure})}{P(\text{event}|\text{no exposure})}. \quad (1)$$

Each case therefore acts as its own control.

In the simplest form of case-crossover design the relative risk of an event under two levels of exposure is assessed. For each case, the exposure at the time of the event is observed, as is also the exposure at a well defined, comparable, previous time. The exposures are binary, often the absence or presence of a stimulant.

A typical example is provided by a study of the association between serious accidental injury and alcohol consumption, conducted by Margie Peden *et al.* [1997], [1999] for the South African Medical Research Council. Here the event is 'accident' suffered by cases observed at a hospital casualty unit, and exposure is binary: having or not having consumed alcohol. The exposure on the 'accident' day and on the previous day was recorded. Table 1 gives a summary of the data.

Table 1: Accident-Alcohol Data

Previous day	Accident day		Total
	Alcohol	No alcohol	
Alcohol	35	16	51
No alcohol	84	81	165
Total	119	97	216

The quantity of interest is the relative risk. In this example the relative risk (1) is

$$\frac{P(\text{Accident} \mid \text{Alcohol})}{P(\text{Accident} \mid \text{No alcohol})} \quad (2)$$

The aim of this paper is briefly to review the rationale underlying the estimation of relative risk from data such as those in Table 1, and to present a hierarchical Bayes analysis of the simplest case-crossover study.

2. Case-distribution Analysis

Greenland [1999] gives a unified approach to the analysis of case-distribution (case-only) studies; the case-crossover study is one of these. Following Greenland we suppose that subject (case) i , (where $i = 1, 2, 3, \dots, n$, the number of cases in the study) can be at certain exposure levels, with probabilities indicated by the following scheme:

Exposure level:	x_1	x_2	$\dots$	x_K
Probability:	p_{i1}	p_{i2}	$\dots$	p_{iK}

(3)

Let D indicate the event. Then, for case i ,

$$P(D \mid x_c) = r_c(i) \cdot \lambda_{0i}(t) \quad \text{for } c = 1, 2, 3, \dots, K$$

where $\lambda_{0i}(t)$ is taken to be a baseline hazard function for case i at time t , and $r_c(i)$ is a 'hazard ratio' for case i associated with exposure level x_c . Note that Greenland's hazard ratio, r_c , does not depend on i . A straightforward calculation gives

$$P(x_c \mid D) = \frac{r_c(i) \lambda_{0i}(t) p_{ic}}{\sum_j r_j(i) \lambda_{0i}(t) p_{ij}} = \frac{r_c(i) p_{ic}}{\sum_j r_j(i) p_{ij}} \quad \text{for } c = 1, 2, 3, \dots, K \quad (4)$$

which does not depend on $\lambda_{0i}(t)$. Substituting a model, such as the loglinear

model of equation 6, for the hazard ratios, $r_j(i)$, will produce a conditional likelihood contribution for each case i . The relative risk can be estimated from the conditional likelihood provided that estimates for the p_{ij} can be found.

In a case-crossover study, the exposure probabilities p_{ij} are computed from the exposure history of the case for some period before its occurrence. Under certain assumptions this leads to setting p_{ij} equal to the proportion of the observed history that case i spent at exposure level x_j .

3. Simple Case-Crossover Analysis

In a simple case-crossover design, the exposures are binary. Let $x = 1$ if case i was exposed and $x = 0$ if not. Then equation 3, simplifies to:

Exposure level x :	1	0
Probability:	p_i	$1 - p_i$

for $i = 1, 2, 3, \dots, n$.

For case i the exposures are recorded at the time of the event t_a and also a well-defined previous time t_b . Let X_{ai} and X_{bi} be the observed exposure levels for case i at times t_a and t_b respectively. It is assumed that the exposure probabilities p_i are time-invariant and therefore X_{ai} and X_{bi} have independent Bernoulli(p_i) distributions.

Therefore, for $x = 0, 1$; $m = a, b$; and $i = 1, 2, 3, \dots, n$

$$P(X_{mi} = x) = p_i^x (1 - p_i)^{1-x},$$

and from the conditional probability in equation 4, we have

$$P(X_{ai} = x | D) = \frac{r_x(i) \cdot p_i^x (1 - p_i)^{1-x}}{r_1(i) \cdot p_i + r_0(i) \cdot (1 - p_i)}. \quad (5)$$

The event D was not observed at time t_b . This implies that

$$P(X_{bi} = x | D) = P(X_{bi} = x) = p_i^x (1 - p_i)^{1-x}.$$

The hazard ratios $r_x(i)$, for case i at exposure x at time t_a (the time of the event), may be modelled in terms of exposure and covariates for each observed case. Suppose that $\mathbf{z}_i$ is a vector of covariate values for case i , γ is a vector of covariate coefficients and β , the log(relative risk) is the coefficient of the exposure x , where $x = 0, 1$.

Substituting the loglinear model

$$r_x(i) = \exp(x\beta + \mathbf{z}_i' \gamma) \quad (6)$$

where appropriate into equation 5, we obtain the following conditional (on D) likelihood contribution for observed exposure x at time t_a from case i

$$\frac{p_i^x (1 - p_i)^{1-x} \exp(x\beta + \mathbf{z}_i' \gamma)}{p_i \exp(\beta + \mathbf{z}_i' \gamma) + (1 - p_i) \exp(\mathbf{z}_i' \gamma)} = \frac{p_i^x (1 - p_i)^{1-x} \exp(x\beta)}{p_i \exp(\beta) + (1 - p_i)}. \quad (7)$$

Like the baseline hazard, the terms $\exp(\mathbf{z}_i' \gamma)$ cancel out. For this model, the covariates $\mathbf{z}_i$ do not provide information about the parameter β .

4. Estimating the Risk Ratio

The log-linear model in equation 6 implies that the ratio

$$\frac{P(D | X_{ai} = 1)}{P(D | X_{ai} = 0)} = \frac{r_1(i) \cdot \lambda_{0i}(t)}{r_0(i) \cdot \lambda_{0i}(t)} = \frac{\exp(\beta + \mathbf{z}_i' \gamma)}{\exp(\mathbf{z}_i' \gamma)} = \exp(\beta) \quad (8)$$

which is the relative risk for exposure to non-exposure. The parameter β is therefore the log(relative risk).

Then the conditional probabilities in equation 5 become

$$P(X_{ai} = 1 | D) = \frac{\exp(\beta + \mathbf{z}_i' \gamma + \ln(p_i))}{\exp(\beta + \mathbf{z}_i' \gamma + \ln(p_i)) + \exp(\mathbf{z}_i' \gamma + \ln(1 - p_i))}$$

$$= \frac{p_i \exp(\beta)}{p_i \exp(\beta) + 1 - p_i}$$

$$P(X_{ai} = 0 | D) = \frac{1 - p_i}{p_i \exp(\beta) + 1 - p_i} \quad (9)$$

Expression 9 can be used to derive a contribution by case i to a conditional likelihood of the observations, x_{ai} . It has the form

$$f(x_{ai} | p_i, \beta) = \left[\frac{p_i \exp(\beta)}{p_i \exp(\beta) + 1 - p_i} \right]^{x_{ai}} \left[\frac{1 - p_i}{p_i \exp(\beta) + 1 - p_i} \right]^{1 - x_{ai}}, \text{ for } x_{ai} = 0, 1 \quad (10)$$

which is a Bernoulli(p_{ai}) probability function, where

$$p_{ai} = \frac{p_i \exp(\beta)}{p_i \exp(\beta) + 1 - p_i} \quad (11)$$

Hence, using the exposure data (X_{ai}) for the time of the event (t_a) for all cases, an estimate of β , and of the relative risk $\exp(\beta)$ may be found. However, no progress can be made unless the p_i values are known, or can be estimated. Even if the p_i are known, no closed form expression for β maximising the likelihood in equation 10 can be found.

5. Estimating the Exposure Probabilities

In the simplest case-crossover design, the exposures are binary and observed on only two occasions for each case.

The conventional method, see Marshall and Jackson [1993] of estimating the relative risk is the ratio of the number of cases where $X_{ai} = 1$ and $X_{bi} = 0$ to the number of cases where $X_{ai} = 0$ and $X_{bi} = 1$. This approximation is generally a good one, provided that $p_i \exp(\beta) + 1 - p_i$ remains relatively

constant from case to case (c.f. Marshall and Jackson [1993]). This approximation fits into Greenland's [1999] unified approach in the following way. If we assume for case i that the p_i are time-invariant, then simply counting the number of occasions where exposure x occurred will provide an estimate of p_i . In our simple case-crossover example where exposure of every case is observed on only two occasions and on only two levels, we can let $W_i = X_{ai} + X_{bi}$ be the number of occasions on which exposure was at level 1. Then p_i is estimated by the relative frequency 0, $\frac{1}{2}$ or 1 corresponding to observed frequencies of 0, 1 and 2 respectively. The likelihood contributions of cases having the same exposure on both occasions will be equal to one, with the result that these cases provide no information about the relative risk $\exp(\beta)$.

The estimates 0, $\frac{1}{2}$ or 1 of individual p_i values seem rather unrealistic, and it is worthwhile to consider an alternative approach to estimating the individual probabilities. Maritz *et al.* [1999] propose an empirical Bayes method for estimating the prior probabilities in an intuitively more realistic way. They assume that the exposure probabilities have a common beta distribution. The parameters of the beta distribution are obtained by matching moments. They match two moments of $W_i = X_{ai} + X_{bi}$ (the number of occasions on which exposure was at level 1), which is binomial($p_i, 2$) with the corresponding moments of the common beta distribution. The exposure probabilities are estimated as the conditional (on the observed exposures) means of the beta distributions

We decided to use a hierarchical Bayes approach in which the distributions of the exposure probabilities and the relative risk are modelled jointly. Our approach is similar to an example in Gelman *et al.* [1995]. The following method was used.

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Biplot applications everywhere: From quality control to stinkwood

N.J. le Roux* & S. Gardner† (*Dept. of Statistics & Actuarial Science,
University of Stellenbosch

†(British American Tobacco

njlr@maties.sun.ac.za

Modern principal component biplots representing various industrial process variables are considered. Together with response data, relationships among variables are graphically displayed. Multivariate process quality is assessed by coloured regions and non-parametrical regression techniques are applied to indicate trends over time in this plot. In a second study the problem is to distinguish between stinkwood and two related woods with appearances very similar to that of stinkwood. In practice decisions are made according to perceptual judgement. A method based on the biplot methodology of Gower & Hand (1996) using wood anatomical characteristics is introduced. After a variable selection technique non-redundant anatomical features are identified and utilized in a canonical variate biplot showing classification regions. Items of unknown origin are interpolated into the biplot. A decision of origin is based on the position of the interpolated item in the biplot. Since distances between points in this biplot are Pythagorean, graphical evidence of the validity of decisions taken, is provided.

A comparison of the referral rates of trainees and trainers: An application of the Generalized Linear Mixed Model.

Jonathan Levin & Ronald Ingle

Unit for Biostatistics, MRC

Dept. of Family Medicine, Medunsa

jonathan.levin@mrc.ac.za

Judicious referral of patients from general practitioner to specialist is an important function. It might be expected that in an academic practice the referral behaviour of the family physician trainer would differ from that of a novice trainee.

In this paper the results from an analysis of 34 047 records of 13 junior registrars, 7 senior registrars and 7 senior physicians are presented. To allow

uninformative. Inference about the parameter of interest, β , can now be made by integrating out the unknown parameters.

The prior distributions used for integrating out the hyperparameters ν_1 and ν_2 , were both $\text{gamma}(2, 10)$. The $\text{gamma}(2, 10)$ distribution has mean 20 and variance 200. This prior distribution provides reasonable support to a wide range of possible values of the parameters. (The analysis was performed for a variety of different sets of parameters for the hyper-prior gamma distributions.) The resulting posterior distribution of β was very similar for the different choices. The prior distribution for β was $\text{normal}(1, 1000)$. The conditional (on D) likelihood is then proportional to:

$$\prod_{i=1}^n \left\{ \nu_1 \exp\left(-\frac{\nu_1}{10}\right) \cdot \nu_2 \exp\left(-\frac{\nu_2}{10}\right) \cdot \frac{1}{\sqrt{2000\pi}} \exp\left(-\frac{(\beta-1)^2}{2000}\right) \right. \\ \cdot \frac{\Gamma(\nu_1)\Gamma(\nu_2)}{\Gamma(\nu_1+\nu_2)} p_i^{\nu_1-1} (1-p_i)^{\nu_2-1} \cdot p_i^{x_{bi}} (1-p_i)^{1-x_{bi}} \cdot p_i^{x_{ai}} (1-p_i)^{1-x_{ai}} \\ \left. \cdot \frac{\exp(x_{ai}\beta)}{p_i \exp(\beta) + 1 - p_i} \right\} \quad (14)$$

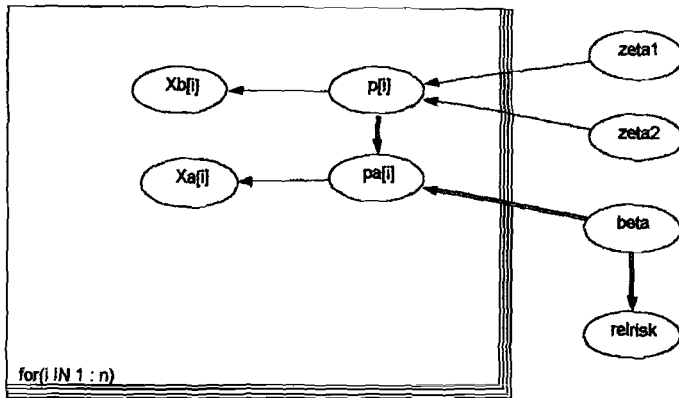
The posterior distributions of the parameters unfortunately cannot be obtained in closed form. Gibbs sampling (a Markov chain Monte Carlo simulation technique) was used to implement the model within a fully Bayesian framework. This is implemented using the WinBUGS program, which provides a means of analysing complex models essentially by describing their structure and automatically deriving the expressions necessary for Gibbs sampling.

7. Using WinBUGS

The freely available software package WinBUGS, provides a means of analysing complex models essentially by describing their structure and automatically deriving the expressions necessary for Gibbs sampling. The basic idea behind the Gibbs sampling algorithm is to successively sample from the conditional distribution of each variable given all the others in the model (these are known as full conditional distributions). More details on Gibbs sampling as implemented in WinBUGS, can be found in Spiegelhalter *et al.* [1996], [1997], [1999]. There are also several interesting examples of Bayes analyses using BUGS in Berry and Stangl [1996].

WinBUGS uses a graphical interface called DoodleBUGS to construct the model and a windows interface for controlling the analysis. There are also graphical tools for monitoring convergence of the simulation. The model specification and simulation are separate components: text versions of the model can be output from DoodleBUGS and the text model descriptions are used as input to the simulation program. Spiegelhalter *et al.* [1999a], [1999b] contain detailed examples.

The model for the simplest case-crossover study is expressed in the form of a *Doodle*. The nodes in the graph denote the data and parameters of the model. The arrows drawn between nodes indicate the conditional independence of the model.



The model is checked by WinBUGS, and if it is structurally correct, WinBUGS generates the code.

The WinBUGS code for the model analysed in this paper, corresponding to the *Doodle* above, is given below. Comments are preceded by # and ignored by WinBUGS.

```

model;
{
  for( i in 1 : n ) {
    p[i] ~ dbeta(zeta1,zeta2)      #exposure probability has beta distribution
    pa[i] <- p[i]*(exp(beta))/(1-p[i]+p[i]*exp(beta))
    Xa[i] ~ dbern(pa[i])           #observed exposure at time of event
    Xb[i] ~ dbern(p[i])            #observed exposure at earlier time
  }
  # prior distributions:
  zeta1 ~ dgamma(2, 0.1)          #note different parametrisation of precision parameter
  zeta2 ~ dgamma(2, 0.1)
  beta ~ dnorm(1, 0.001)         #note different parametrisation of precision parameter
  relrisk <- exp(beta)            #relative risk = exp(beta)
}

```

The code is only the definition of the model. WinBUGS constructs the appropriate likelihood function and then simulates corresponding posterior distributions. The data can be specified in various forms. The data for the simplest case-crossover study consists of two columns ($Xa[]$ and $Xb[]$) of digits, 0 or 1. There is a row for each case i and the digit indicates the exposure of the case at the relevant time. The WinBUGS program, together with detailed instruction manuals containing worked examples and comments (c.f. Spiegelhalter *et al.* [1996], [1997], [1999], [1999a], [1999b]) may be obtained free of charge (at the moment) from the BUGS website: <http://www.mrc-bsu.cam.ac.uk/bugs/mainpage.html>.

8. Analysis of Accident-Alcohol Data

Refer to the 'accident-alcohol' data in Table 1. In the conventional analysis described in section 5, see Marshall and Jackson [1993], the cases with estimated p_i equal to 0 or 1 contribute a factor 1 to the likelihood so that it is

$$L(\beta) = \left(\frac{\exp(\beta)}{\exp(\beta) + 1} \right)^{84} \left(\frac{1}{\exp(\beta) + 1} \right)^{16},$$

maximized by $\exp(\hat{\beta}) = 84/16 = 5.25$, or $\hat{\beta} = 1.658$. The 95% confidence limits are set by treating the observed frequency 84 conditionally as a realisation of a Binomial random variable with index parameter $n = 84 + 16 = 100$.

For the hierarchical Bayes analysis of section 5, inference on the parameters of interest, $\exp(\beta)$ and the exposure probabilities, p_i , are based on the posterior marginal distributions, which were obtained using the simulation technique known as Gibbs sampling. It can be shown that under broad conditions this process eventually provides samples from the joint posterior distribution of the unknown quantities. Empirical summary statistics were formed from these samples and used to draw inferences about their true values. Although a different exposure probability p_i was modelled for each case i , the probabilities converged to only three different distributions, say θ_{00} , θ_{diff} and θ_{11} , corresponding to non-exposure at both occasions (0,0), different exposures on the two occasions (0,1 or 1,0) and exposure at both occasions (1,1), respectively. The posterior distributions of the exposure probabilities and the relative risk, $\exp(\beta)$, are all positively skewed. The means of the distributions, with the 95% confidence interval for the relative risk are given (labelled HB) in the summary Table 2 below. The values are based on a

sample of 11000 simulations, following a burn-in of some 800 simulations, for which the values were excluded for the statistics. Estimates obtained by the conventional method and by Maritz *et al.* [1999] (labelled EB, for empirical Bayes) are included.

Table 2: Accident-Alcohol Estimates

Method	θ_{00}	θ_{diff}	θ_{11}	$\exp(\beta)$	95% confidence interval	
Conventional	0	0.5	1	5.25	3.17	10.11
HB	0.213	0.248	0.283	4.28	2.77	6.45
EB	0.371	0.400	0.429	1.89	1.45	2.47

9. Cellphones and Collisions

Redelmeier & Tibshirani [1997] analyse a study in which the 'cases' are drivers involved in a motor vehicle collision (the event) and the two levels of exposure are using or not using a cellular telephone while driving. Whether or not the driver was using a cellular telephone was noted at the time of the collision and also at the equivalent time on the day preceding the collision. Their study involves an extensive collection of covariates, which are not relevant to our study. The data are given in Table 3.

Table 3: Cellphone Data

Previous day	Collision day		Total
	Cellphone	No Cellphone	
Cellphone	13	24	37
No Cellphone	157	505	662
Total	170	529	699

Following the conventional approach we find that the maximum likelihood estimate of β is given by $\exp(\hat{\beta}) = 157/24 = 6.54$ or $\hat{\beta} = 1.878$. The means of the posterior distributions of the exposure probabilities and the relative risk

and the 95% confidence interval for the relative risk from the hierarchical Bayes analysis given in Table 4. The corresponding conventional and EB estimates of Maritz *et al.* [1999] are included.

Table 4: Cellphone Estimates

Method	θ_{00}	θ_{diff}	θ_{11}	$\exp(\beta)$	95% confidence interval	
Conventional	0	0.5	1	6.54	4.32	11.07
HB	0.048	0.072	0.094	6.32	4.16	9.61
EB	0.148	0.148	0.148	1.85	1.56	2.20

10. Evaluating the Bayes Model

The hierarchical Bayes model for each of the two datasets was evaluated by calculating the expected frequency of cases (out of the total number of cases observed) in each cell, using equation 12, with estimates of all the parameters. These estimated frequencies were compared to the observed frequencies (given in Tables 1 and 3) by calculating the χ^2 goodness-of-fit statistics. The calculated χ^2 -values were all smaller than 2, which indicates that the models fit the observed data satisfactorily. Expected frequencies for the empirical Bayes model of Maritz *et al.* [1999] resulted in larger (greater than 50) χ^2 goodness-of-fit statistics. This indicates that the models do not fit the observed data satisfactorily.

The point estimates for the hierarchical Bayes relative risk varied by less than 10% from the reported values for different parameter sets of the hyper-prior gamma distributions having the same coefficient of variation as the reported distributions.

The posterior distributions of the parameters β and the p_i are all positively skewed, and it was found that a better fit, as indicated by a smaller χ^2 -

value is obtained when using the mode, rather than the mean of the posterior distributions as estimates of the parameters. The mode was approximated by subtracting the distance between the mean and the median from the median ($\text{mode} \approx \text{median} - (\text{mean} - \text{median})$). This approximation was decided on after visually examining the density traces of the posterior distributions generated by the WinBUGS software. A more sophisticated approach was not deemed necessary.

11. Discussion

The analysis of case-crossover studies is straightforward using the unified approach presented by Greenland [1999], provided that information on the 'usual' exposure for each case is available for constructing the exposure probabilities. Marshall and Jackson [1993] say: *Without this information a case cross-over analysis cannot be undertaken, since the analysis compares behaviour at the time of the event with the usual pattern of behaviour.* That is why the simplest case-crossover design analysed in this paper cannot be analysed in a totally satisfactory manner using conventional methods. This is because the conventional analysis regards those cases with constant exposure as noninformative with respect to the exposure probabilities. Part of the data is therefore ignored.

A hierarchical Bayesian approach, in which the distributions of the exposure probabilities and the relative risk are modelled jointly, produces parameter estimates that fit the data of the examples reasonably well. The model is implemented using the WinBUGS program, which provides a means of analysing complex models essentially by describing their structure and automatically deriving the expressions necessary for Gibbs sampling. The basic

idea behind the Gibbs sampling algorithm is to successively sample from the conditional distribution of each variable given all the others in the model (these are known as *full conditional distributions*).

In this paper specific numerical values were used for the parameters of the hyper-prior gamma distributions. In fact, the same pairs of values were used for both beta parameters. These values were selected because they provide reasonable support to a wide range of possible values of the parameters of the prior beta distribution of the exposure probabilities for the two examples discussed here. Should our technique be used for other examples, one would probably use different parameters for the hyper-prior gamma distributions.

In the examples, the Bayes estimates of the relative risk (summarised by the means) are substantially smaller than the conventional estimates. The estimates obtained by Maritz *et al.* [1999] are even more substantially smaller than the hierarchical Bayes estimates.

Examination of the likelihood function shows that, whether or not the Bayes estimates are smaller than the conventional estimates depends on the values of the exposure probabilities, which of course, depend on the observed frequencies. In the examples the frequencies in the cells corresponding to no exposure on both occasions are relatively large. This makes the estimated exposure probabilities of those cases relatively small and a corresponding reduction in the estimate of risk ratio intuitively plausible. This argument applies to both the hierarchical Bayes estimates and the empirical Bayes estimates of Maritz *et al.* [1999].

The estimated standard error of $\hat{\beta}$ under the Bayes scheme is considerably smaller than that obtained by the conventional approach, which is shown in the width of the confidence intervals in tables 2 and 4.

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